

Chedam — Spec v0.3: Decisions and New Features

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This version records the owner decisions made on the Reality Check and Gap List, adds eight new feature specifications, and recommends how tax and returns should work. It builds on v0.1 and v0.2; the Master Specification combines all three.

1. Summary of decisions

Chedam stays offline-first and uses the internet only when the store chooses to connect, for payments, owner remote view, email fallback and updates. Features are grouped into modules the owner turns on, so small stores see only what they use.

#	Topic	Decision	Section
D1	Simplicity	Core always on; extra features are modules the owner switches on; a shop-type preset pre-selects them	2
D2	Card payments	Now: standalone Square (or similar) terminal, which can take payments offline. Later: Square Terminal API when the hub has internet, including through a tablet or phone hotspot	3
D3	Owner away from the shop	Offline app by default; when internet is available, the hub uploads an encrypted copy to the owner's Google Drive or OneDrive, and the Chedam app reads it remotely (read-only)	4
D4	Data quality	Required fields, decimal and unit rules; a product cannot be sold until complete	5
D5	Promotions	Product and category promotions from a list of types; system calculates new prices	6
D6	Labels	Batch label printing by scan, category or list; price changes and promotions auto-add items to the batch; Letter and A4, rows × columns	7
D7	Loyalty	Points per \$ set by owner; customer found by phone number or printed loyalty card; no emails or calls	8

#	Topic	Decision	Section
D8	Notifications	In-app first; email to personal address only if unread and the hub is online	9
D9	Updates	Tax rates, software and architecture updates download when the hub is online	10
D10	Tax and returns	Recommendation in Sections 11–12, decision pending	11, 12
D11	Temperature monitoring	Later, as a new Monitoring menu	13
D12	Support	Owner (Sreya) provides support; plan developed as the product matures	—

2. Feature on/off: core plus switchable modules

Features are grouped like apps on a phone: the core is always on, and the owner switches on extra modules during setup or later in Settings → Features. A switched-off module disappears from menus, forms, reports and the dashboard, so staff never see it.

Modules

Module	Contains	Always on?
Core: Sell	Sales terminal, tills, receipts, returns, cash rounding, holds, training mode	Yes
Core: Stock	Products, packs, stock levels, receiving, damaged/loss, phone scan, bulk add	Yes
Core: Team	Users, roles, messages, pings, notifications, tasks	Yes
Promotions	Promotions, coupons, scheduled prices, near-expiry markdowns	Switch
Labels	Shelf and price label batches	Switch
Customers & Loyalty	Customer profiles, points, store credit, feedback	Switch
Orders & Layaway	Layaway, special orders, quotes, reserved client orders	Switch

Module	Contains	Always on?
Vendors & Purchasing	Vendors, purchase orders, auto-PO, bill photo reading, price lists, consignment	Switch
Delivery Apps	Uber Eats, DoorDash, SkipTheDishes orders	Switch
Weighed Goods	Scale, PLU codes, price per kg/lb	Switch
Regulated Items	Age checks, container deposits, eco fees	Switch
Product Variety	Variants (size/colour), bundles, serial numbers	Switch
Expenses	Staff out-of-pocket claims, petty cash	Switch
Time & Schedules	Clock in/out, schedules, breaks	Switch
HR & Payroll	Employee records, payroll preparation, ROE/T4 data	Switch
Finance	Reconciliation, bank deposits, tax returns, statements	Switch
Operations	Opening/closing checklists, handover notes, licence reminders	Switch
Connected Services	Card terminal link, owner remote view, email fallback, delivery auto	Switch (needs internet)
Monitoring	Fridge/freezer sensors and logs	Later

Shop-type presets

Preset	Modules switched on
Grocery / perishables	Promotions, Labels, Customers & Loyalty, Vendors & Purchasing, Weighed Goods, Regulated Items, Operations
Convenience store	Promotions, Labels, Regulated Items, Vendors & Purchasing, Operations
Clothing / gifts	Promotions, Labels, Customers & Loyalty, Product Variety, Orders & Layaway
Food takeout	Delivery Apps, Promotions, Customers & Loyalty, Operations

Preset	Modules switched on
General retail	Promotions, Labels, Customers & Loyalty, Vendors & Purchasing

Rules

- Only the owner can switch modules on or off.
- Switching off never deletes data; switching back on restores everything.
- A module that another depends on cannot be switched off while in use (e.g. Labels needs nothing, but Delivery Apps auto mode needs Connected Services).
- The setup wizard shows the preset as a checklist with one-line descriptions, so the owner understands each choice.

3. Card payments

Chedam records card payments taken on the store's own terminal (Square or similar) now, and links to the terminal directly later, when the hub has internet. Customers are told plainly: Chedam is an offline app; card payments run on your payment terminal, which needs its own connection or a phone/tablet hotspot.

Stage 1 — standalone terminal (ships in Phase 1)

1. Cashier finishes the cart and taps **Card (terminal)**.
2. Chedam shows the exact amount in large type; cashier keys it into the terminal.
3. After approval, cashier taps **Approved** and enters the last 4 digits or the terminal's receipt number (optional setting).
4. Declined → cashier taps **Declined**; the sale stays open for another payment.
5. End of day: the terminal's settlement report total is entered (or imported as CSV from Square) and Reconciliation matches it to Chedam's card sales.

The terminal's own offline mode (where the processor offers it) lets the store keep taking cards when the internet is down; those payments are approved later by the bank and can be declined, which is the processor's risk rule, not Chedam's.

Stage 2 — linked terminal (Connected Services module, Phase 5)

- **Square Terminal API:** Chedam's hub sends the amount to Square's cloud, which pushes it to the paired Square Terminal; the result comes back to the sale automatically. This requires internet on the hub and device pairing with the merchant's Square account ([Square](#)).

- The hub checks payment status by asking Square (polling), so no inbound connection to the shop is needed.
- **Internet source:** the store's router, or a tablet/phone acting as a hotspot that the hub joins.
- **No internet at checkout:** Chedam falls back to Stage 1 on that sale automatically, with a banner "Terminal link offline — key amount manually".
- **Other processors:** a common payment interface so Helcim, Stripe Terminal, Moneris or others can be added later; the business chooses by cost.

What Chedam stores

Payment method, amount, terminal/processor, approval reference, last 4 digits (never full card numbers), status. No card data passes through Chedam, so card-security certification (PCI) stays with the processor.

4. Owner remote view

Yes, this is possible. When the hub has internet, it uploads an encrypted copy of the store's key figures to the owner's own Google Drive or OneDrive; the owner opens Chedam on their phone anywhere and it reads that copy. It is read-only and only as fresh as the last upload.

How it works

1. **Setup (once, owner only):** Settings → Connected Services → Remote view → choose Google Drive or OneDrive → sign in and allow Chedam to use one private folder ("Chedam-StoreName"). Chedam asks only for access to files it creates, not the whole drive.
2. **Uploads:** while online, the hub uploads every 15 minutes (setting): today's sales, payments by method, top items, stock alerts, open tasks, till status, and the end-of-day report. A full encrypted backup goes up nightly.
3. **Encryption:** files are encrypted on the hub with a key only the owner's Chedam app holds. Google or Microsoft see unreadable files.
4. **Owner's phone:** the owner's Chedam app (installed on the phone while it was on the shop Wi-Fi) shows a **Remote** badge when away from the shop, signs in to the drive, downloads the latest file and shows the dashboard: "Last updated 2:45 pm".
5. **No internet at the shop:** nothing uploads; the phone shows the last copy with its time, so the owner knows it is old.

Limits

- Read-only: the owner cannot change prices or approve refunds remotely in this version.
- Freshness depends on the shop's internet.
- Manager access to remote view is optional and uses a second key the owner can revoke.
- Drive storage counts against the owner's own quota (a year of summaries is a few MB; nightly backups are larger and keep the last 14).

5. Data validation rules

Every product has a status: **Draft** (saved but not sellable) or **Active** (sellable). A product becomes Active only when all required fields for its type are valid; the till cannot sell a Draft product, and a task lists every Draft waiting to be finished.

Required fields by product type

Product type	Required to become Active
Every product	Name, category, selling unit, price, tax class, at least one of barcode / PLU / till button
Sold by weight	Unit (kg or lb), price per unit, PLU or scale code, tare weight (can be 0), "approved scale required" acknowledged
Pack / case	Contains quantity and the unit it contains
Perishable	Shelf life (days) or expiry required at receiving, storage area
Age-restricted	Minimum age (18 or 19 in BC) and restriction type
With deposit / eco fee	Deposit or fee amount and its tax treatment
Imported	HS code, country of origin

Number rules

Field	Rule
Money (price, cost, fees)	Stored in cents; exactly 2 decimals shown; no negatives; price ≥ 0.01 for sellable items

Field	Rule
Weight	Up to 3 decimals (0.742 kg); must be > 0 on a sale; no negatives
Quantity, "each" items	Whole numbers only; > 0 on sales and receiving
Quantity, weighed items	Up to 3 decimals
Percent (discount, tax)	0-100, up to 3 decimals
Stock adjustments	Negative allowed only with a reason code (damage, loss, count correction)

Units

- Weighed products choose **kg or lb** and keep it; the till can display both.
- Conversion: 1 lb = 0.45359237 kg exactly; converted values rounded only at display, never in storage.
- Price per kg and per lb shown together on labels when the store chooses (common for produce).

Validation behaviour

- Errors appear beside the field as you type; the Save button says what is missing ("2 fields needed").
- **Save as draft** is always allowed so no work is lost.
- Imports apply the same rules (v0.1 Section 12); failing rows become Drafts with the reason listed.
- Changing a rule (e.g. making expiry required) re-checks existing products and lists those that no longer pass, without blocking sales of already-active items until the owner chooses.

6. Promotions engine

The owner picks a promotion type from a list, chooses products or a category, enters X and Y, and Chedam shows the new effective price and margin for every affected product before saving. At the till, promotions apply automatically.

Promotion types

Type	Owner enters	Example	Till behaviour
% off	X %	20% off all Snacks	Line price – 20%
\$ off	\$X	\$1 off Mango Pulp	Line price – \$1
Fixed sale price	\$X	Basmati 5 kg now \$14.99	Line price = \$14.99
Buy Y, get X% off	Y items, X %	Buy 3, get 15% off each	When qty ≥ 3, each – 15%
Buy Y, get \$X off	Y items, \$X	Buy 2, save \$1.50	\$1.50 off per set of 2
Multi-buy price	Y items for \$X	3 for \$5	Every 3 cost \$5; extras at normal price
Buy Y get Z free	Y, Z	Buy 2 get 1 free	Cheapest item in the set free
Mix and match	Y items from a group for \$X or X% off	Any 4 juices for \$10	Any products in the group count
Spend threshold	Spend \$Y, get \$X or X% off	Spend \$50, get \$5 off	Applied to cart total
Time-based	Any type + days and hours	10% off bakery after 6 pm	Active only in that window
Near-expiry markdown	X% by days left	30% off 2 days before expiry; 50% on last day	Applied to that lot only; label printed
Coupon	Any type + code	Code SUMMER10	Applied when code scanned or typed

Setting up a promotion

1. Promotions → **New** → pick type from the list.
2. Choose scope: products (scan or search), a category, or several categories; optional exclusions.
3. Enter X and Y, start and end date/time, optional days and hours.
4. **Preview:** table of affected products with regular price, promo price, cost, new margin %; rows below cost shown red with a warning.
5. Save → affected products are added to the label batch (Section 7).

Rules

- **No stacking by default:** if several promotions match, the customer gets the single best one. Owner can mark a promotion "stacks with others" (e.g. coupons).
- Priority order when tied: item-level before category-level before cart-level.
- Limits: per transaction, per customer (needs loyalty), total quantity.
- Receipt shows each saving line by line and total saved.
- Returns refund the price actually paid after the discount was shared across the qualifying items (Section 12).
- Promotions end automatically; labels for ended promotions are queued for reprint at regular price.
- Reports: sales, units, margin and savings per promotion vs the same days before it.

7. Label printing

Labels are printed in batches. Items reach the batch three ways: the user scans them, picks a category, or selects from the product list; and Chedam adds items automatically whenever their price changes or a promotion starts or ends. Printing the batch clears it.

Adding items to the batch

- **Scan:** Labels → Scan; each scan adds 1 label (scan again for more, or set quantity).
- **Category:** pick one or more categories; all active products added.
- **List:** search and tick products from the full list.
- **Automatic:** price change, cost-driven price update, promotion start/end, near-expiry markdown, new product activated. Each auto-added item shows its reason ("Price changed \$3.49 → \$3.99").
- Duplicates merge; the latest price wins.

Label content (template editor)

Name (2 lines max), price (large), unit price (per kg/lb or per 100 g/ml), barcode, promotion text and end date ("Sale until Oct 20"), regular price struck through, store logo (optional), date printed, PLU, origin (optional), bilingual name (optional).

Paper layouts

Layout	Paper	Grid (columns × rows)	Use
Address-label sheet	US Letter	3 × 10 (30 per sheet)	Standard shelf labels
Large label sheet	US Letter	2 × 5 (10 per sheet)	Promotion signs
A4 label sheet	A4	3 × 8 (24 per sheet)	Shelf labels
Plain A4 / Letter, cut lines	A4 or Letter	User sets columns × rows	Cut by hand and place
Custom	Either	User sets columns, rows, margins, gaps	Any label brand
Label printer roll	Thermal 2" or 3"	1 per label	If a label printer is added

Printing

1. Labels → Batch shows count by reason and a sheet preview.
2. **Start at position:** choose which label on a partly used sheet to start from, so no labels are wasted.
3. **Print** creates a PDF sized exactly to the sheet and opens the normal print dialog (works with any printer).
4. After printing, the user confirms "Printed OK" and the batch clears; "Reprint" stays available for the last 10 batches.
5. Test page with outlines to check alignment once per label brand.

8. Loyalty program

Customers earn points on what they spend and redeem them as a discount at the till. The owner sets the earn rate and the value of a point. Customers are found by phone number or by a printed loyalty card; Chedam never emails or calls them.

Owner settings

Setting	Example default
Earn rate	1 point per \$1 spent (before tax, after discounts)
Point value	100 points = \$1 off

Setting	Example default
Minimum to redeem	500 points
Excluded items	Tobacco, lottery, gift-type items, deposits
Earn on promotions	Yes / No
Points expiry	Never, or after 12 months without a visit
Bonus rules (optional)	Double points on a category or day

At the till

1. Cashier taps **Customer** and types the phone number or scans the loyalty card.
2. Not found → **Join** in 10 seconds: first name and phone (or card number only). Customer agrees verbally; the cashier ticks "Customer agreed".
3. Screen shows points balance and dollar value: "1,240 pts = \$12.40".
4. **Redeem** asks how many points (up to the sale total); applied as a discount line before tax.
5. Receipt prints points earned, redeemed and new balance.

Loyalty cards

- Card numbers are generated by Chedam; the store prints them on card stock using the label feature (business-card layout) or orders plastic cards with the barcode.
- A card can be linked to a phone number later; lost cards are blocked and points moved to a new card.

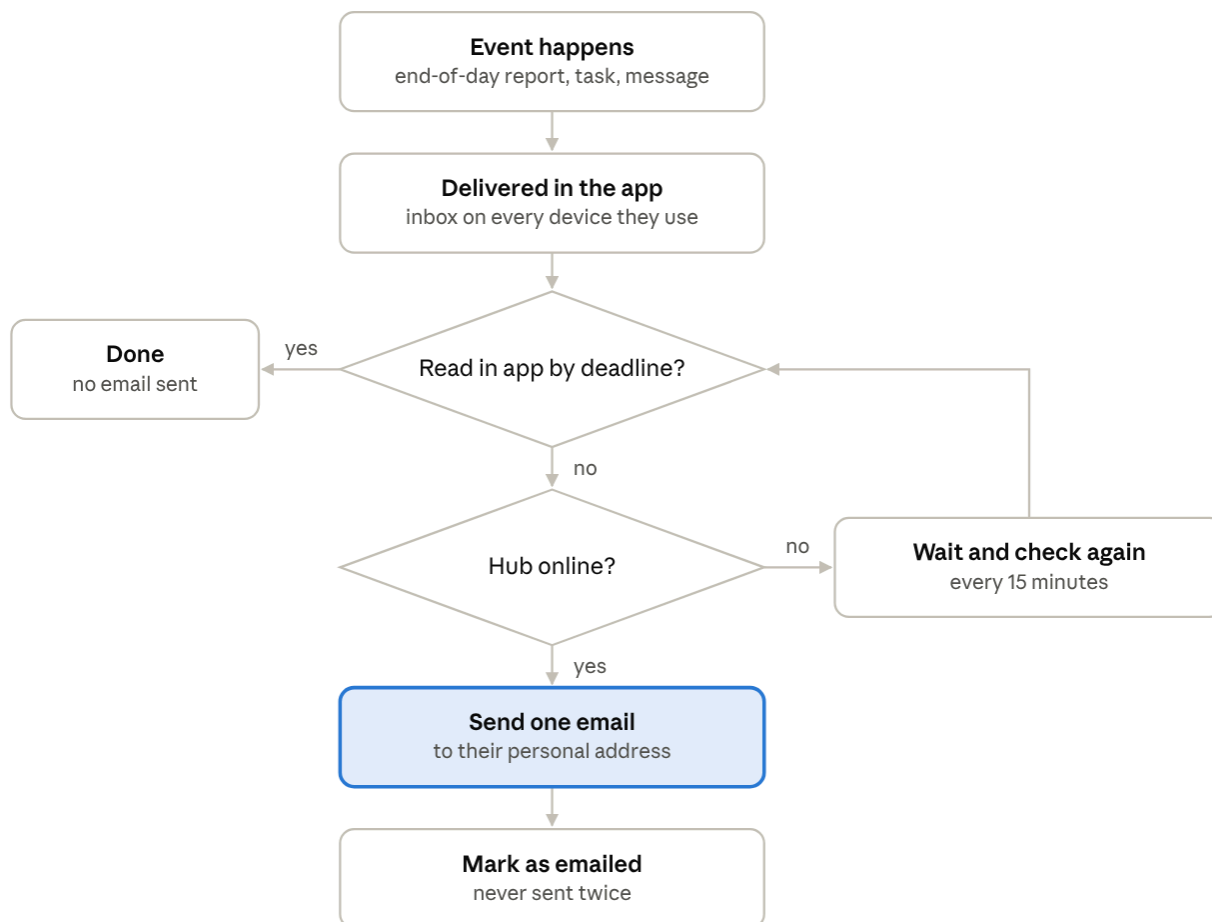
Rules

- Returns reverse points earned on the returned items; points redeemed on a returned sale are given back.
- Offline tills earn and redeem against cached balances; conflicts (same points used twice) are flagged as a task.
- Customer data: first name, phone and/or card number, purchase history only. Customers can ask to see or delete their data (BC PIPA).
- Reports: members, active this month, points outstanding (a liability shown in \$), top customers.

9. In-app notifications with email fallback

Every report, task and message goes to the person's Chedam inbox first. Only if it is still unread at its deadline, and the hub has internet, does Chedam send one email to the person's personal address. A message read in the app is never emailed.

Email goes out only when a message is unread and the hub is online



notification delivery · in-app first, email fallback

The wait loop re-checks the read status, so a message read in the app while the hub is offline is never emailed later.

Details

- **Deadline per type** (owner setting): end-of-day report — next morning 8 am; urgent tasks (backup failed, till not closed) — 30 minutes; normal messages — 24 hours; some types never email (pings).
- **Who gets what:** owner and managers choose which report types they receive; staff receive only their own messages and schedule changes.
- **Read** means opened in the app on any device; the read status syncs through the hub.

- **Sending account:** the owner connects one email account (Gmail or Outlook) in Connected Services; Chedam sends through it, so no paid email service is needed.
- **Email content:** a short summary and "Open in Chedam for details"; no sensitive pay or HR details in email.
- Each email is logged (to whom, when, which message); a message is emailed at most once.

10. Updates when the hub is online

Whenever the hub has internet, it checks Chedam's update channel and downloads signed update packages; the store keeps running on the current version until the owner (or the schedule) installs them. USB updates remain for stores that never connect.

What updates

Package	Contents	Install
Tax tables	GST/HST/PST rates and effective dates by province, deposit and eco-fee schedules	Automatic; new rates switch on their effective date; owner notified
App	New features, fixes, security patches	Owner approves, or auto at night if set
Architecture / schema	Database migrations, hub system changes	Always backup first; owner approves; automatic rollback if checks fail
Datasets	HS codes, product data, shelf-life tables	Background, during quiet hours
Hub OS	Security updates for Raspberry Pi OS	Weekly, at night, with reboot after backup

Rules

- Every package is signed by Chedam; the hub refuses unsigned or altered files.
- Updates never install during trading hours (from business settings).
- The health page shows current versions, last check, and pending updates.
- Tax changes show the owner what will change and when (e.g. which tax class changes, old and new rate, effective date).

- Hosting the update channel: a free static host (e.g. GitHub Releases) to keep running costs at \$0.

11. Recommendation: tax handling

Recommended: every product carries a **tax class**, the tax rates live in an updatable table (not in code), tax is calculated per tax type on the whole receipt, and every sale stores the exact tax it charged so returns and reports never recalculate history. Confirm the class list and treatments with an accountant before Phase 1.

Tax classes (BC starting set)

Tax class	GST 5%	BC PST 7%	Typical items (to confirm with accountant)
Standard	Yes	Yes	Most general goods
GST only	Yes	No	Items exempt from PST but subject to GST, e.g. many prepared foods
Zero-rated	0%	No	Basic groceries (fruit, vegetables, rice, flour, milk)
Exempt	No	No	Items outside GST/PST
Custom	Set by owner	Set by owner	Anything the accountant advises

Other provinces (HST) are added as rows in the same table, so the design works across Canada.

How tax is calculated

1. Each line's price after promotions is grouped by tax type.
2. Tax per type = group total × rate, rounded to the cent once per type per receipt (not per line), to avoid penny drift.
3. Loyalty redemptions and cart discounts are spread across lines in proportion to their price before tax is calculated.
4. Deposits and eco fees use their own tax class (treatment confirmed by accountant).
5. Cash rounding to 5¢ happens after tax, on the final cash total only.

Prices shown with or without tax

Business setting: **tax added at till** (normal in Canada) or **tax included in price** (some markets, food stalls). Labels and receipts follow the setting.

Tax-exempt sales

- Cashier taps **Tax exempt**, picks a reason (First Nations status card, PST resale/wholesale, other) and records the required number; manager approval optional.
- Exempt sales appear in a separate report with the recorded reference numbers for audits.

Reports

Tax collected and refunded by tax type and rate, for any period, laid out to match the GST/HST return and the BC PST return; exempt and zero-rated sales shown separately.

12. Recommendation: returns, damage, loss, breaks and timestamps

Recommended: a return always refunds exactly what the customer paid on the original sale (same price, same discount share, same tax), never today's price; every stock change and every staff action carries a reason, a person, a device and a timestamp.

Return policy settings (owner)

Setting	Recommended default
Return window	30 days; per-category override (perishables: same day or not returnable)
Non-returnable flag	Per product or category (e.g. cut fruit, lottery, opened food)
With receipt	Refund to original payment method
Without receipt	Manager approval; store credit only, at the lowest price in the last 30 days
Restocking fee	Off; optional % per category
Refund limit without manager	\$50

Return rules

1. Find the sale by scanning the receipt barcode, or search by date, amount, card last 4 or loyalty customer.
2. Select lines and quantities; Chedam blocks returning more than was sold (including earlier returns).
3. Refund amount = original line price after its share of promotions and cart discounts, plus the tax originally charged on it.
4. Loyalty points earned on returned items are removed; points redeemed are given back proportionally.
5. Card refunds: in Stage 1 the cashier refunds on the terminal and confirms; in Stage 2 the linked terminal does it.
6. **Exchange** = return + new sale on one screen; only the difference is paid or refunded.
7. Each returned item gets a stock decision: back to stock, damaged, return to vendor, or dispose.

Damage and loss

Reason code	When	Stock effect	Cost goes to
Damaged in store	Dropped, broken	– qty	Shrink: damage
Spoiled / expired	Perishable past date	– qty	Shrink: spoilage
Customer return — damaged	From returns	Not restocked	Shrink: returns
Vendor return	Sent back for credit	– qty, credit expected	Vendor credit
Theft / unknown loss	Found missing at count	– qty	Shrink: unknown
Count correction	Count found more or less	± qty	Inventory adjustment
Staff use / samples	Tasting, store use	– qty	Expense: samples

Every entry: product, lot, quantity, reason, optional photo, person, device, timestamp.
Adjustments over a set value need manager approval.

Breaks and timestamps

- Staff tap **Start break / End break** on any device; the hub records the time from its own clock (devices' clocks are ignored for time-keeping).
- BC rule check: a 30-minute meal break within every 5 consecutive hours worked; warnings shown to the employee and manager before it is missed.
- Edits to punches or breaks keep the original time, the new time, who changed it and why.
- Every record in Chedam stores created-at, updated-at, by whom and on which device (v0.1 event log); times are stored in UTC and shown in the store's time zone.

13. Monitoring menu (later)

A future **Monitoring** menu will watch fridges, freezers and other storage with small wireless temperature sensors that report to the hub over the shop network, so perishables are protected without anyone writing logs by hand. Not in the first release.

- Each sensor is linked to a storage area (Cooler 1, Freezer, Display fridge) and its allowed range.
- Live view per area: current temperature, last 24 hours chart, door-open spikes.
- Out-of-range for a set time → urgent ping to all tills and the manager, and a task listing the products and lots stored there.
- Manual log option for stores without sensors (twice-daily entry).
- Inspector-ready report by date range.
- Hardware to evaluate later: low-cost Wi-Fi or Zigbee sensors that can work without any cloud account.

14. Gap list decisions applied

These are the Decision column values from the Reality Check and Gap List, now part of the plan.

Add

- **Checkout:** cash rounding to 5¢; split payments; price override with reason; no-sale drawer log; training mode; tax-exempt sales; hold and recall across tills; customer-facing display; store credit and house accounts; layaway and special orders; quotes; foreign cash (USD).

- **Products and pricing:** promotions engine; coupons; scheduled prices; near-expiry markdowns; shelf labels; produce PLU and certified scale; container deposits and eco fees; age-restricted products; product images on till; variants; bundles; serial/IMEI tracking.
- **Inventory:** recall trace; shrink report; min/max auto purchase orders; consignment stock; bilingual labels; supplier price lists.
- **Customers:** customer profiles; loyalty; feedback portal (in-store tablet or QR when online).
- **Staff and operations:** opening/closing checklists; shift handover notes; loss-prevention report; staff discount rules; break and overtime alerts; licence and permit reminders; accessibility.
- **Finance and compliance:** GST/HST receipt rules; GST and PST return reports; 6-year record retention; ROE data; T4 data; petty cash; bank deposit tracking; card fee tracking; financial statements; sales insights; end-of-day report (in-app, email fallback).

Later

Integrated card terminal (Stage 2), fridge temperature log and sensors (Monitoring), cycle counts, multi-location, online store, Google Business Profile, customer pre-orders, commissions, equipment maintenance log, payroll service connection.

Skip

Gift cards, email/SMS receipts, marketing consent and campaigns, CCTV link, multi-language interface.

Not yet decided

Gap list G (support line, remote support, hardware kit, installation, warranty, terms, insurance, pricing model) — to be shaped with the support plan.

15. Open questions

- ☐ Tax: confirm tax classes, deposit/eco-fee treatment and zero-rated items with an accountant.
- ☐ Returns: accept the recommended defaults (30 days, \$50 cashier limit, store credit without receipt)?
- ☐ Payments: which processor for Stage 2 first — Square, or compare Helcim/Stripe on cost?
- ☐ Remote view: upload every 15 minutes OK, or only at end of day to save data?

- ☐ Email fallback: which email account sends (owner's Gmail/Outlook), and how long to wait before emailing (default: next morning 8 am)?
- ☐ Loyalty: default earn rate and point value for the pilot store?